# The Adaptation Spectrum: RAG → Fine-tuning → Grounding

## 1. Lightest option first

✎ DIAGRAM




Cost, effort and risk rise as you move right. Stop at the first rung that meets the need.

The adaptation spectrum.

- **In-context learning (ICL)**: instructions and examples in the prompt; no training run
- **RAG**: inject external knowledge at run time ('the model doesn't know our facts')
- **Grounding**: link outputs to sources for trust and auditability
- **PEFT** (LoRA, adapters): tune a small slice of weights for style, format or terminology. Cheap, fast, low-risk
- **Full fine-tuning**: retrain substantial weights for deep specialisation. Costly, easy to overdo
- Rule: **RAG for facts, fine-tuning for behaviour and style**

## 2. The Agentic AI Maturity spectrum

1 **Basic**: single agent, fixed workflow, predefined tools

2 **Dynamic tools**: chooses from a set of approved tools per problem

3 **Introspective**: **ReAct** (alternate reasoning and acting) + **Reflexion** (self-critique and correction)

4 **Multi-agent**: specialists coordinate in supervised, auditable workflows

5–6 **Collaborative**: decentralised negotiation, consensus, swarm

- Each stage brings new failure modes, so each calls for new patterns: dynamic tools need routing, introspection needs self-correction, multi-agent needs coordination and conflict resolution

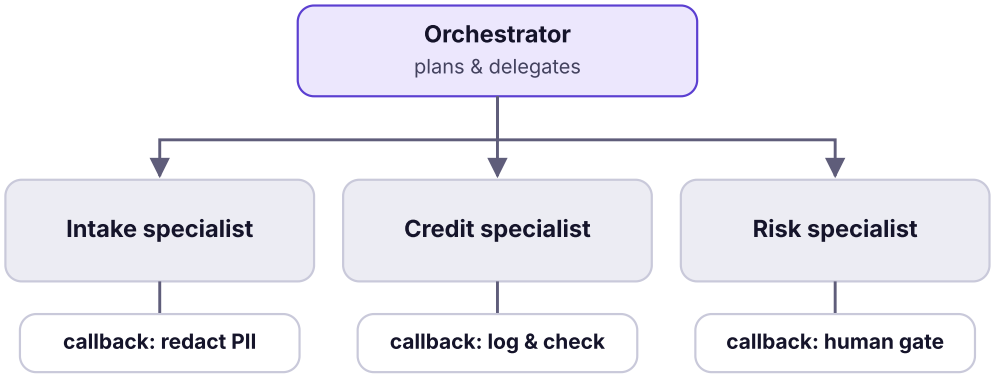
## 3. RAG as sensing

- Support agent → latest product and policy docs · analyst agent → current market data · compliance agent → current regulations
- For agents, retrieval is **perception**: how the agent sees the current state of its domain instead of relying on frozen, generic weights

### 4. Hierarchical architecture

- An **orchestrator** decomposes the goal and delegates; **specialists** each own one sub-task with their own tools; results flow back up
- **Callbacks** wrap every hand-off: hooks that inspect, redact, filter or block input and output. This is where governance lives: PII redaction, safety checks, human-in-the-loop gates, the audit trail

✎ DIAGRAM




Specialists do the work; callbacks on every hand-off make the hierarchy controllable.

SAY IT

*"I'd reach for the lightest adaptation that works: prompting, then RAG, then grounding, then PEFT, and full fine-tuning last."*